

ZECHARIAH TAN

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EDUCATION

- National University of Singapore** Aug. 2026 – Present
Bachelor of Computing in Computer Science
- Temasek Polytechnic** Apr. 2021 – May 2024
Diploma in Information Technology with Merit 3.98 GPA
- Course Valedictorian
 - AWS Course Bronze Medal
 - DBS Major Project Prize
 - SCS Skills Mastery Award

EXPERIENCE

- WorldSkills Web Technologies Competitor & Coach** Apr. 2024 – Sep. 2024
Temasek Polytechnic Singapore
- Earned an *International Bronze Medal* in Web Technologies representing Singapore at WorldSkills Lyon 2024
 - Developed training materials and coached competitors for WorldSkills Singapore 2025
 - Coached a competitor who earned a *Gold Medal* at WorldSkills Singapore and WorldSkills ASEAN 2025
- Software Engineer Intern** Oct. 2023 – Feb. 2024
Centre for Strategic Infocomm Technologies (CSIT) Singapore
- Built 3 Spring Boot microservices and a Next.js dashboard for indexing and viewing document data
 - Integrated search, storage, message queues, and database services across the microservice architecture
 - Deployed the microservices to Kubernetes and built observability pipelines for monitoring and logging
 - Developed unit and integration tests for backend services as project requirements evolved

PROJECTS

- YTMusic API** | *TypeScript, npm* Dec. 2021 – Jul. 2024
- Published a type-safe YouTube Music API wrapper for retrieving songs, artists, lyrics and search results
 - Grew the package to 2,500+ weekly npm downloads, 160+ GitHub stars and 40+ forks
 - Built runtime validation and automated tests to handle inconsistent upstream API responses
- Finpoint** | *Laravel, PHP, React, TypeScript, Sqlite* Apr. 2026 – Present
- Built a personal finance tracker that consolidates transactions across multiple banks
 - Categorised imported transactions into financial records for clearer spending analysis
- TheRook** | *Rust* Aug. 2025 – Sep. 2025
- Built a chess engine in Rust using bitboards and precomputed attack data for efficient legal move generation
 - Implemented Stockfish-backed perf tests to validate move generation across search trees reaching 15M+ nodes
- SounDroid V2** | *Flutter, Dart, Express, TypeScript, Firebase* Apr. 2022 – Aug. 2022
- Built a full-stack mobile music streaming app with synchronised lyrics, offline downloads, and queue management
 - Built user flows for authentication, account management, playlists, and search history

TECHNICAL SKILLS

Languages: TypeScript, Rust, Java, PHP, Dart, SQL
Frameworks: React, Next.js, Flutter, Laravel, Spring Boot, Express, Tailwind CSS
Databases: Elasticsearch, MySQL, PostgreSQL, Sqlite, Firestore
Cloud & Hosting: AWS, Vercel, Firebase, Cloudflare, Fly.io, Heroku
Infrastructure & DevOps: Docker, Kubernetes, NGINX, GitHub Actions, Prometheus, Grafana